

The Recycling Robot Example from Sutton & Barto

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1 The problem statement

This is the running example from Sutton & Barto (Chapter 3, Section 3.1). A mobile robot has the job of collecting empty soda cans in an office environment. It has sensors to detect cans and a battery. The robot's battery can be in one of two states:

- **high** (fully charged),
- **low** (partially discharged).

The robot can perform three actions:

- **search** — actively look for cans,
- **wait** — stay still and wait for a can,
- **recharge** — go to the charging station.

Rewards:

- +1 if the robot finds a can (or is handed one),
- 0 if it finds nothing or recharges,
- -3 if it runs out of battery and must be rescued.

The question is: what is the optimal policy?

2 The MDP in Sutton & Barto notation

In Sutton & Barto notation, the dynamics are given by the joint distribution

$$p(s', r \mid s, a) = \Pr\{S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a\}.$$

States.

$$\mathcal{S} = \{\text{high}, \text{low}\}.$$

Actions.

$$\mathcal{A} = \{\text{search}, \text{wait}, \text{recharge}\}.$$

Discounted case. We take $\gamma < 1$. The undiscounted (average-reward) case is discussed in Section ??.

Table 1: Dynamics $p(s', r | s, a)$ for the recycling robot.

s	a	s'	r	$p(s', r s, a)$
high	search	low	+1	α
high	search	low	0	$1 - \alpha$
low	search	low	+1	β
low	search	high	-3	$1 - \beta$
high	wait	high	+1	1
low	wait	low	+1	1
high	recharge	high	0	1
low	recharge	high	0	1

3 The dynamics table

The canonical dynamics for the recycling robot are given in the following table. Each row specifies the probability of a transition (s', r) from state s under action a .

Interpretation.

- **search from high:** the battery always goes to low. The reward is +1 with probability α (found a can) and 0 with probability $1 - \alpha$ (found nothing).
- **search from low:** with probability β , the robot finds a can (reward +1) and the battery stays low. With probability $1 - \beta$, the robot runs out of battery, must be rescued (reward -3), and the battery goes to high.
- **wait:** reward +1, battery unchanged.
- **recharge:** reward 0, battery goes to high.

4 Bellman equations for a fixed policy

Suppose the policy is

$$\pi(\text{search} | \text{high}) = 1, \quad \pi(\text{search} | \text{low}) = 1.$$

The Bellman evaluation equations are

$$v_\pi(s) = \sum_{s', r} p(s', r | s, \pi(s)) [r + \gamma v_\pi(s')].$$

Substituting the table:

$$\begin{aligned} v_\pi(\text{high}) &= \alpha[1 + \gamma v_\pi(\text{low})] + (1 - \alpha)[0 + \gamma v_\pi(\text{low})] = \alpha + \gamma v_\pi(\text{low}), \\ v_\pi(\text{low}) &= \beta[1 + \gamma v_\pi(\text{low})] + (1 - \beta)[-3 + \gamma v_\pi(\text{high})]. \end{aligned}$$

Solving. From the first equation,

$$v_\pi(\text{high}) = \alpha + \gamma v_\pi(\text{low}).$$

Substituting into the second:

$$v_\pi(\text{low}) = \beta + \beta\gamma v_\pi(\text{low}) + (1 - \beta)(-3) + (1 - \beta)\gamma[\alpha + \gamma v_\pi(\text{low})].$$

Collecting terms in $v_\pi(\text{low})$:

$$v_\pi(\text{low})[1 - \beta\gamma - (1 - \beta)\gamma^2] = \beta - 3(1 - \beta) + (1 - \beta)\gamma\alpha.$$

Hence

$$\begin{aligned} v_\pi(\text{low}) &= \frac{\beta - 3(1 - \beta) + (1 - \beta)\gamma\alpha}{1 - \beta\gamma - (1 - \beta)\gamma^2}, \\ v_\pi(\text{high}) &= \alpha + \gamma v_\pi(\text{low}). \end{aligned}$$

Numerical example. Take $\alpha = 0.5$, $\beta = 0.5$, $\gamma = 0.9$. Then

$$v_{\pi}(\text{low}) = \frac{0.5 - 1.5 + 0.225}{1 - 0.45 - 0.405} = \frac{-0.775}{0.145} \approx -5.34,$$

$$v_{\pi}(\text{high}) = 0.5 + 0.9(-5.34) \approx -4.31.$$

Both values are negative: always searching from a low battery is a bad policy because of frequent -3 rescues.

5 Optimal Bellman equations

The optimality equations are

$$v_*(\text{high}) = \max \left\{ \begin{array}{l} \text{search: } \alpha + \gamma v_*(\text{low}), \\ \text{wait: } 1 + \gamma v_*(\text{high}), \\ \text{recharge: } 0 + \gamma v_*(\text{high}) \end{array} \right\},$$

$$v_*(\text{low}) = \max \left\{ \begin{array}{l} \text{search: } \beta + \beta \gamma v_*(\text{low}) + (1 - \beta)[-3 + \gamma v_*(\text{high})], \\ \text{wait: } 1 + \gamma v_*(\text{low}), \\ \text{recharge: } 0 + \gamma v_*(\text{high}) \end{array} \right\}.$$

6 Solving the optimality equations

For $\alpha = \beta = 0.5$, $\gamma = 0.9$, let us test the policy “search when high, recharge when low”.

Policy evaluation.

$$v(\text{high}) = 0.5 + 0.9 v(\text{low}), \quad v(\text{low}) = 0.9 v(\text{high}).$$

Substituting:

$$v(\text{high}) = 0.5 + 0.81 v(\text{high}) \implies v(\text{high}) = \frac{0.5}{0.19} \approx 2.63,$$

$$v(\text{low}) = 0.9 \cdot 2.63 \approx 2.37.$$

Check argmax from high.

- search: $0.5 + 0.9 \cdot 2.37 = 2.63$.
- wait: $1 + 0.9 \cdot 2.63 = 3.37$. **Larger!**
- recharge: $0.9 \cdot 2.63 = 2.37$.

So **wait** is better than search from high.

Check argmax from low.

- search: $0.5 + 0.5 \cdot 0.9 \cdot 2.37 + 0.5[-3 + 0.9 \cdot 2.63] = 1.26$.
- wait: $1 + 0.9 \cdot 2.37 = 3.13$. **Larger!**
- recharge: $0.9 \cdot 2.63 = 2.37$.

So **wait** is better than recharge from low.

Optimal policy. The optimal policy is “wait always”, with

$$v_*(\text{high}) = v_*(\text{low}) = \frac{1}{1 - \gamma} = 10.$$

7 Fixed point perspective

The optimality operator is

$$(\mathcal{T}_*v)(s) = \max_a \sum_{s',r} p(s',r \mid s,a) [r + \gamma v(s')].$$

It is a γ -contraction in the sup norm:

$$\|\mathcal{T}_*u - \mathcal{T}_*v\|_\infty \leq \gamma \|u - v\|_\infty.$$

Theorem 1 (Banach). *Let (X, d) be a complete metric space and $T : X \rightarrow X$ a contraction with factor $\gamma < 1$. Then T has a unique fixed point x_* , and for any x_0 , the sequence $x_{n+1} = Tx_n$ converges to x_* with*

$$d(x_n, x_*) \leq \gamma^n d(x_0, x_*).$$

Applying Banach to $\mathcal{T}_*$ on $(\mathbb{R}^2, \|\cdot\|_\infty)$: there is a unique v_* with $\mathcal{T}_*v_* = v_*$, and value iteration

$$v_{k+1}(s) = \max_a \sum_{s',r} p(s',r \mid s,a) [r + \gamma v_k(s')]$$

converges to v_* geometrically.

Value iteration for $\alpha = \beta = 0.5$, $\gamma = 0.9$. Start with $v_0(\text{high}) = v_0(\text{low}) = 0$.

Table 2: Value iteration for the recycling robot.

k	$v_k(\text{high})$	$v_k(\text{low})$	greedy action
0	0	0	—
1	1	1	wait
2	1.9	1.9	wait
3	2.71	2.71	wait
$\vdots$	$\vdots$	$\vdots$	$\vdots$
∞	10	10	wait

The values approach 10 geometrically, and the optimal action is always **wait**.

8 Other parameter regimes

The optimal policy depends on α , β , and γ . For example:

- If α is large (searching from high is very likely to find a can) and β is small (searching from low is likely to run out), the optimal policy is “search when high, recharge when low”.
- If β is large (searching from low is safe), the optimal policy may be “search always”.
- If γ is close to 1 (far-sighted), waiting becomes very attractive because it yields a steady stream of rewards.

The thresholds can be derived by comparing the values of the candidate policies at the argmax.

9 Average-reward (undiscounted) case

In the original S&B presentation, the recycling robot is discussed in the **average-reward** setting, where $\gamma = 1$ and one maximizes the long-run average reward per step:

$$r(\pi) = \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right].$$

The optimality equations become

$$h_*(s) + r_* = \max_a \sum_{s', r} p(s', r \mid s, a) [r + h_*(s')],$$

where h_* is the **bias** (relative value) function and r_* is the optimal average reward. The Banach contraction argument no longer applies directly because $\gamma = 1$; instead, one uses the theory of ergodic Markov chains and the span seminorm.

For the recycling robot with $\alpha = \beta = 0.5$, the average reward of “wait always” is 1 per step, and this turns out to be optimal.

Summary

Concept	Recycling robot
States	high, low
Actions	search, wait, recharge
Rewards	+1 (can), 0 (nothing/recharge), −3 (rescue)
Key parameters	α, β, γ
Dynamics	Table ??
Bellman evaluation	Linear system in $v_\pi(\text{high}), v_\pi(\text{low})$
Bellman optimality	max over actions
Optimal policy	Depends on α, β, γ
Example ($\alpha = \beta = 0.5, \gamma = 0.9$)	wait always, $v_* = 10$
Fixed point	$v_* = \mathcal{T}_* v_*$, unique by Banach